

University of Central Punjab

COMPUTER COMMUNICATIONS AND NETWORKS

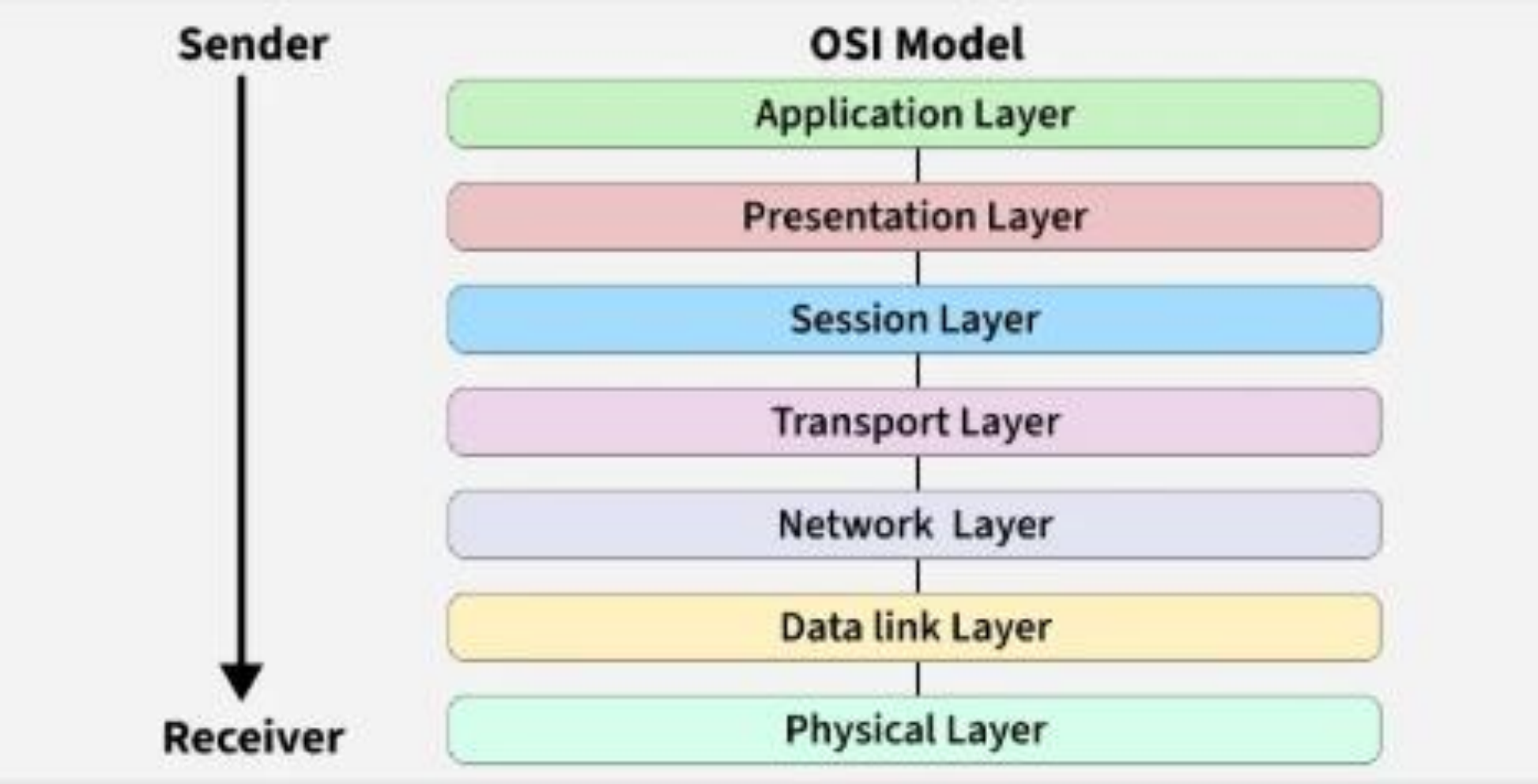
(CSNC-2413)

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OSI Model (Open System Interconnection)

- The **OSI** Model is a conceptual framework created by the International Organization for Standardization **(ISO)** to describe how data is transmitted across a network using a structured seven-layer architecture.
- Divides network communication into seven functional layers.
- Assigns specific responsibilities to each layer.
- Promotes compatibility between different networking systems.
- Simplifies network design, implementation, and troubleshooting.

OSI Model (Open System Interconnection)



The Application Layer

- he **Application Layer** sits at the top of the OSI stack and serves as the direct interface between the software user and the network. It produces the data that will be sent and displays the information received from other layers.

The Application Layer

- **User Interface:** Acts as a "window" for network-based applications (like web browsers or email clients) to access network services.
- **Core Protocols:** Utilizes high-level protocols such as **HTTP/S** (Web), **SMTP** (Email), **FTP** (File Transfer), and **DNS** (Domain Name Resolution).

The Application Layer



The Presentation Layer

- Often called the **Translation Layer**, the **Presentation Layer** ensures that data is formatted, secured, and compressed so that the receiving application can correctly interpret it.
- **Translation:** It extracts data from the Application Layer and converts it into a standardized format for network transmission, bridging differences in data representation between different systems.

The Presentation Layer

- **Encryption/Decryption:** Provides security by converting "plain text" into "cipher text" (encryption) and back again (decryption) using key values. This process is typically handled by protocols like **TLS/SSL**.
- **Compression:** Reduces the total number of bits required for transmission, which increases network efficiency and speed.

The Presentation Layer



The Session Layer

- The **Session Layer** acts as the "dialogue manager," governing the opening, closing, and security of communication channels between two devices.
- **Session Lifecycle:** Manages the establishment, maintenance, and termination of connections between applications.
- **Authentication & Security:** Ensures that the communicating parties are verified and the connection is secure.

The Session Layer

- **Dialog Control:** Directs whether communication is half-duplex (alternating) or full-duplex (simultaneous).
- **Practical Example:** In a web-based messenger, the Session Layer maintains the active link between your browser and the server, ensuring your specific chat remains open and synchronized while handling background encryption and data conversion.

The Session Layer



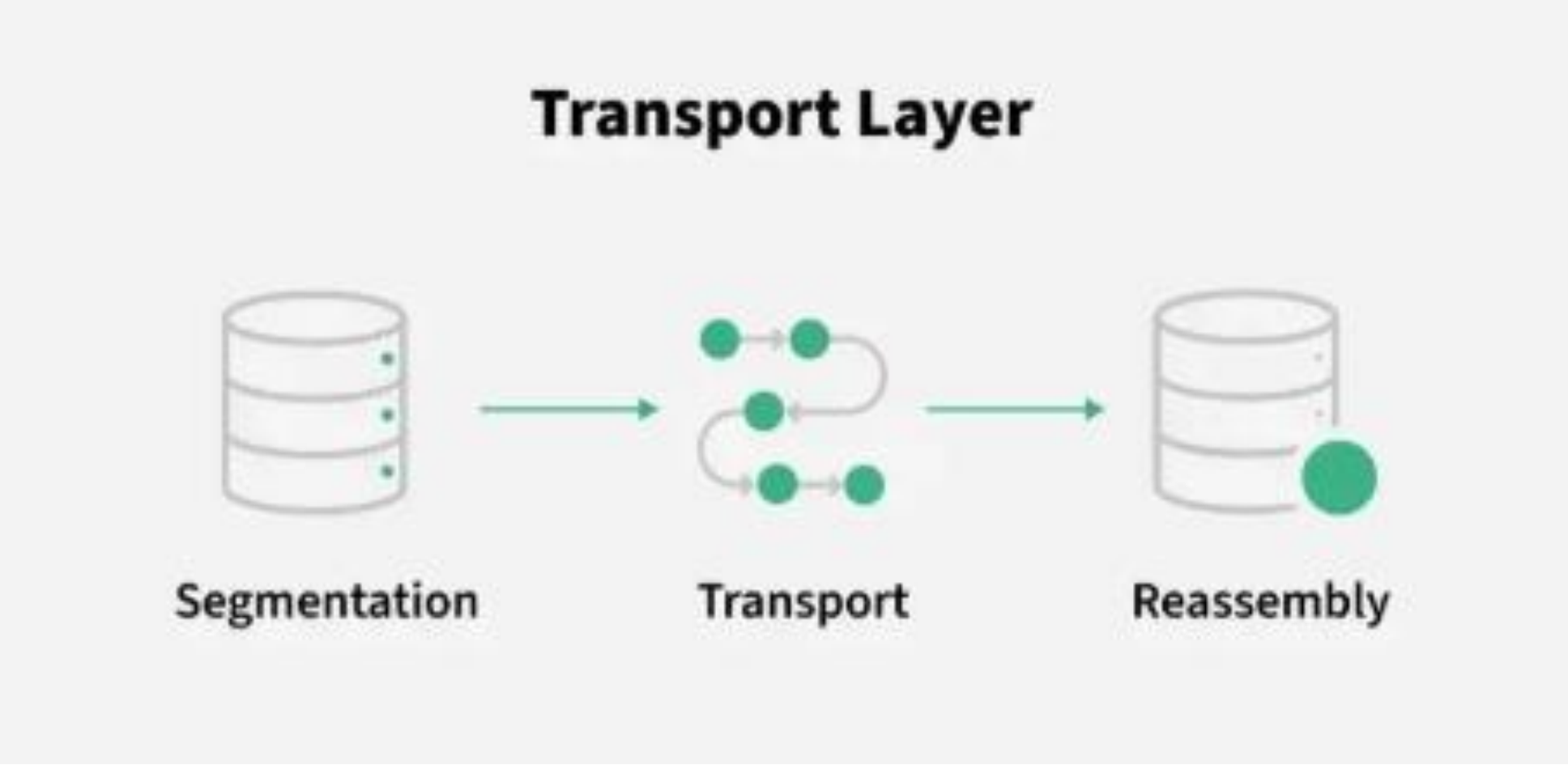
The Transport Layer

- The **Transport Layer** ensures the end-to-end delivery of entire messages. It acts as a liaison, providing services to the Application layer while utilizing the infrastructure of the Network layer to ensure data reaches the correct application on the destination host.
- **Data Unit:** Data is broken down into Segments.
- **Service Point Addressing:** Uses Port Numbers (e.g., Port 80 for web traffic) to ensure data is delivered to the specific process or application intended, not just the device.

The Transport Layer

- **Segmentation & Reassembly:** Splits large messages into smaller segments for transmission and reassembles them in the correct order at the destination.
- **Protocols:** Common protocols include **TCP** (reliable), **UDP** (fast).
- **Connection-Oriented (TCP):** Requires a "handshake" to establish a connection; ensures reliability via error checking and acknowledgements.
- **Connectionless (UDP):** Sends data immediately without a formal connection; faster but offers no guarantee of delivery.

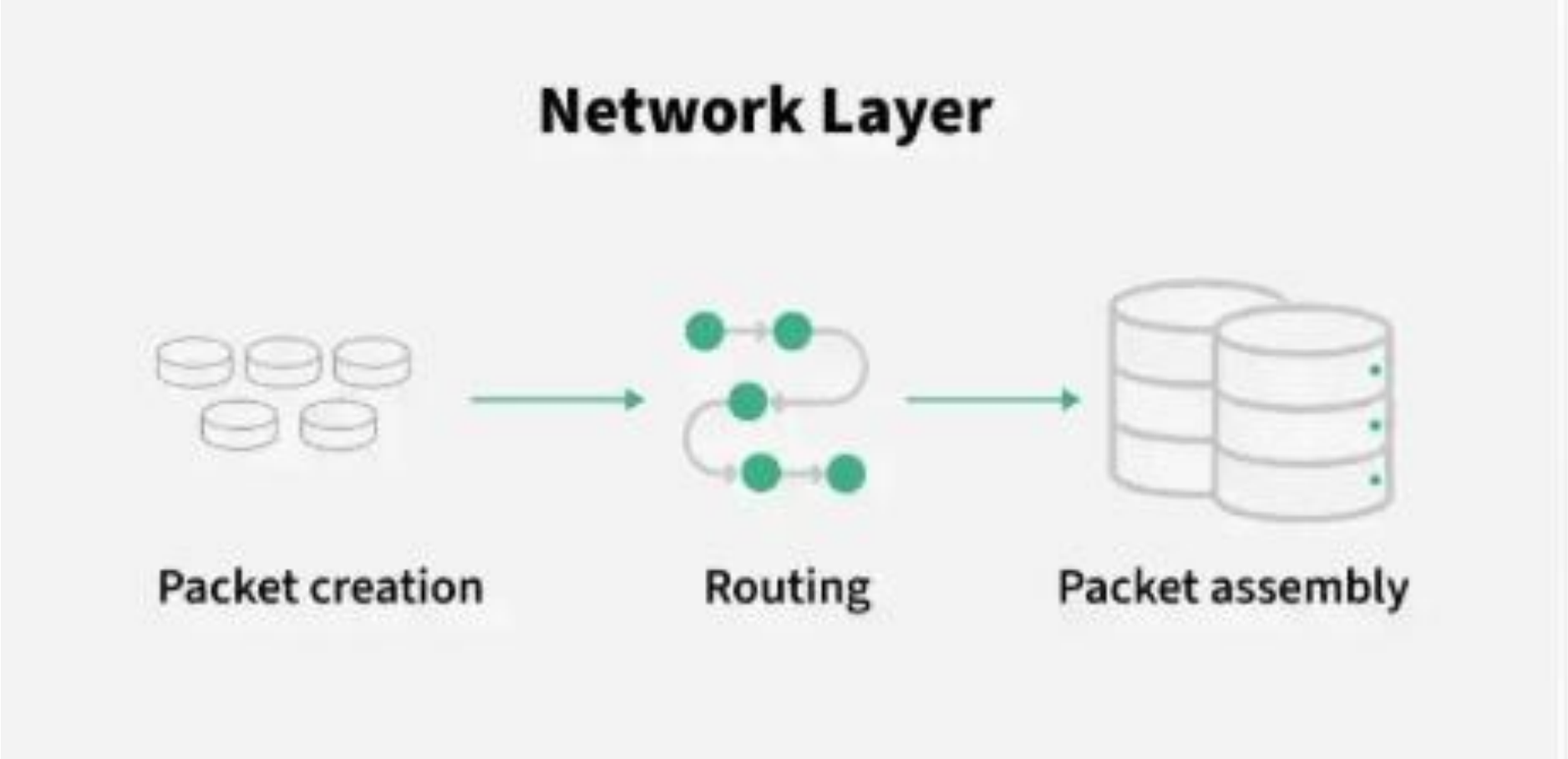
The Transport Layer



The Network Layer

- The **Network Layer** manages data transmission between hosts across different networks by handling logical addressing and path finding.
- **Data Unit:** Data segments are encapsulated into Packets.
- **Logical Addressing:** Assigns unique **IP addresses** (sender and receiver) to the packet header to identify devices globally.
- **Routing:** Determines the most efficient physical path from the source to the destination across interconnected networks.
- **Hardware:** Primarily implemented via **Routers and Switches**.

The Network Layer



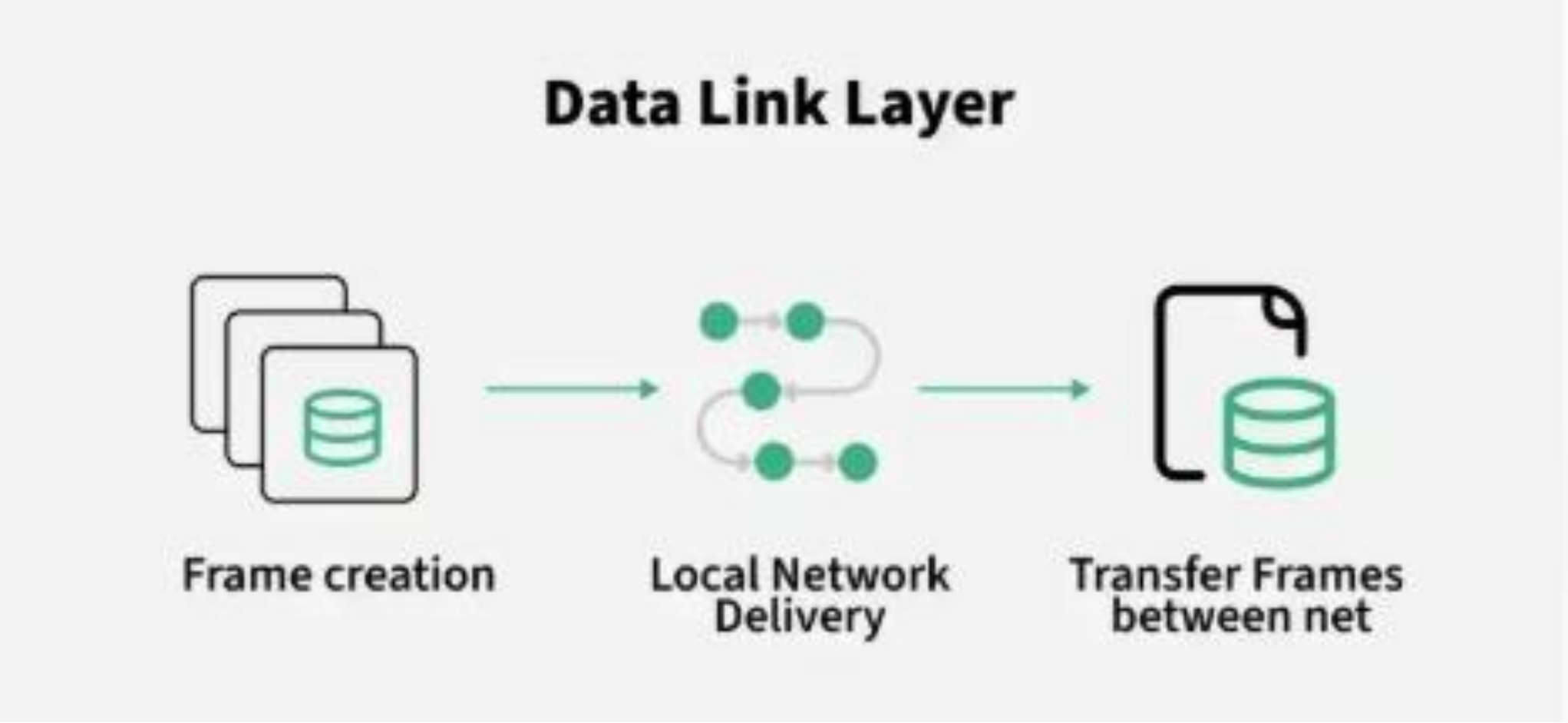
The Data Link Layer

- The **Data Link Layer** serves as the bridge between the physical hardware and the logical network, ensuring reliable node-to-node delivery of data. Its primary goal is to ensure that data transfer is error-free across the physical medium.
- **Data Packaging (Framing):** At this layer, data packets received from the Network layer are divided into manageable units called Frames.
- **Addressing and Hardware:** The DLL uses **MAC addresses** (Physical Addressing) to identify hosts. It encapsulates the sender's and receiver's MAC addresses into the frame header. Common devices operating here are **Switches and Bridges**.

The Data Link Layer

- **Error Control:** It detects and retransmits damaged or lost frames to maintain data integrity.
- **Flow Control:** It synchronizes the data rate between a fast sender and a slower receiver to prevent data "bottlenecks" or loss.
- **Access Control:** When multiple devices share the same communication channel, the MAC sublayer dictates which device has the right to transmit at any given time to avoid collisions.

The Data Link Layer



The Physical Layer

- The **Physical Layer** is the foundation of the OSI model, acting as the bridge for actual physical connections between devices. Its primary mission is the transmission of raw, unstructured bit streams over a physical medium from one node to the next.
- **Core Responsibility:** It manages the hardware-level transmission of individual bits. When receiving data, it translates incoming physical signals back into digital 0s and 1s to be processed by the Data Link layer.
- **Essential Hardware:** Common devices operating at this level include **Hubs**, **Repeaters**, **Modems**, and **Cables**.

The Physical Layer

- **Physical Topologies:** It defines the structural layout of the network, such as Bus, Star, or Mesh configurations.
- **Transmission Mode:** It determines the direction of data flow, utilizing modes like Simplex (one-way), Half-Duplex (two-way, one at a time), or Full-Duplex (simultaneous two-way).

The Physical Layer

